Janghyun **Noh**

AI Researcher

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Education

Myongji University
Seoul, S.Korea

M.S. in Artificial Intelligence — *Sep. 2024 - Aug. 2026*

- Conducted research at the Artificial Intelligence and Data Analytics Laboratory under the supervision of Prof. Minsik Oh.
- Conducted collaborative research with external institutions including the Korea Disease Control and Prevention Agency (KDCA), Seoul National University Hospital (SNUH), Kyungpook National University (KNU), and AIgenDrug Co., Ltd.
- Served as a teaching assistant for undergraduate courses including Deep Learning, Machine Learning, Data Mining, and Basic Programming (Fall 2024 – Spring 2026).
- Received the Research Honor Award from the President of Myongji University upon graduation.

Myongji University
Seoul, S.Korea

B.S. in Data Technology, School of Convergence Software — *Mar. 2018 - Aug. 2024*

- Conducted undergraduate research at the Artificial Intelligence and Data Analytics Laboratory under the supervision of Prof. Minsik Oh.
- Conducted collaborative research on clinical and genetic factors associated with hypertension progression in patients with hyperlipidemia in collaboration with the Korea Disease Control and Prevention Agency (KDCA).

RED School – Alternative School
Geumsan, S.Korea

Student — *Mar. 2012 - Jan. 2018*

- Served as Student Council President.
- Received academic excellence scholarships for two consecutive years.

Experience

Artificial Intelligence and Data Analytics Lab, Myongji University
Seoul, S.Korea

Graduate Researcher — *Sep. 2024 - Aug. 2026*

- Developed scCap, a deep learning framework that predicts patient phenotypes from large-scale single-cell data without manual annotation (first-author paper in Briefings in Bioinformatics; open-source code and Docker image).
- Built a multimodal deep learning model combining clinical records and chest X-rays to predict COVID-19 severity, designed to extend to new modalities (with KDCA).
- Analyzed spatial gene-expression data from breast cancer tissue to characterize cell types and the tumor microenvironment (with SNUH).
- Set up a large-scale genome analysis workflow for apple sequencing data using the open-source nf-core/sarek pipeline (with KNU).
- Developed AI prediction models for drug target discovery using single-cell data (with AIgenDrug).
- Compared gene-expression patterns in liver cancer tissue with and without cirrhosis using spatial data.
- Built an LLM-based AI agent environment (Biomni) for automated genomic data analysis.

Artificial Intelligence and Data Analytics Lab, Myongji University
Seoul, S.Korea

Undergraduate Research Intern — *Dec. 2022 - Aug. 2024*

- Developed machine learning models to predict in-hospital mortality of ARDS patients from ICU records (MIMIC-III), comparing missing-data handling strategies.
- Designed labeling criteria from blood tests and blood pressure, and built machine learning models to predict hypertension onset on large-scale cohort data (with KDCA).

Bigleader
Seoul, S.Korea

AI Intern (Youth AI Internship) — *Jul. 2022 - Oct. 2022*

- Developed a customs notice analysis and monitoring system (with KOTRA).
- Received the KOTRA President's Merit Award and the TriBIG Project Scholarship.

Publications & Presentations

Journal Articles

Annotation-free phenotype prediction using knowledge-augmented clustering from single-cell RNA sequencing data

Briefings in Bioinformatics (BIB)

Published

2026

J. Noh, Y. Shin, M. Kim, and M. Oh

Ferroptosis-based therapeutic strategy targeting iron metabolism and SLC7A11 overcomes oxaliplatin resistance in patient-derived pancreatic cancer organoids

Journal of Experimental & Clinical Cancer Research (JECCR)

Published

2026

J. Choi, K. Lee, M. Oh, **J. Noh**, Y. Kim, E. Hwang, I. Cho, W. Paik, J. Ryu, J. Ku, and S. Lee

Conference Presentations

Patient-Level Metacell Construction with Pre-trained Embeddings for Phenotype Prediction from scRNA-seq

KIISE Korea Computer Congress (KCC)

Oral Presentation

2026

Y. Shin, **J. Noh**, and M. Oh

Lightweight Analysis of Patient Phenotype Prediction Based on Single-Cell RNA-seq Data Reduction Techniques

KIISE Korea Software Congress (KSC)

Poster Presentation (Best Poster Award)

2025

Y. Shin, **J. Noh**, and M. Oh

An annotation-free framework for phenotype prediction using hierarchical multiple instance learning with single-cell RNA-sequencing data

International Conference on Genome Informatics ISCB-Asia

Poster Presentation

2025

J. Noh, Y. Shin, M. Kim, and M. Oh

Patient Phenotype Prediction from Single-cell RNA-seq Data through Annotation-free Multiple Instance Learning

BIOINFO 2025, Korean Society for Bioinformatics (KSBI)

Poster Presentation (Excellent Poster Award)

2025

J. Noh, Y. Shin, M. Kim, and M. Oh

AMIL: Automated Cell Grouping Framework for Phenotype Prediction with Multiple Instance Learning

KIISE Korea Computer Congress (KCC)

Oral Presentation (Best Presentation Award)

2025

J. Noh, M. Kim, and M. Oh

Predicting In-Hospital Mortality in Patients with Acute Respiratory Distress Syndrome Using MIMIC-III Data: A Comparative Study of Machine Learning Models with Varied Missing Data Handling Strategies

KIISE Korea Software Congress (KSC)

Poster Presentation

2023

J. Noh, G. Choi, and M. Oh

Honors & Awards

Awards

2026	**Research Honor Award**, Myongji University	*Seoul, S.Korea*
2025	**Best Poster Award**, KIISE Korea Software Congress (KSC)	*Seoul, S.Korea*
2025	**Excellent Poster Award**, BIOINFO 2025, Korean Society for Bioinformatics (KSBI)	*Seoul, S.Korea*
2025	**Best Presentation Award**, KIISE Korea Computer Congress (KCC)	*Seoul, S.Korea*
2022	**President's Merit Award**, Korea Trade-Investment Promotion Agency (KOTRA)	*Seoul, S.Korea*

Scholarships

2022	**TriBIG Project Scholarship**, Korea Trade-Investment Promotion Agency (KOTRA)	*Seoul, S.Korea*
2019	**Globalization Scholarship**, Myongji University	*Seoul, S.Korea*
2018	**Scholarship III**, Myongji University	*Seoul, S.Korea*
2016	**Academic Excellence Scholarship**, RED School	*Geumsan, S.Korea*
2015	**Academic Excellence Scholarship**, RED School	*Geumsan, S.Korea*

Projects

2024 – 2026	AI-based target discovery platform for CAR-Treg cell therapy in chronic inflammatory disease
2024 – 2026	Genetic and physiological mechanisms of apple blossom dormancy release under climate change
2024 – 2026	Spatial transcriptomics-based target discovery and validation for refractory solid tumors
2025 – 2026	Client-side web platform for detecting large CRISPR-Cas9-induced DNA deletions
2025 – 2026	GeoMx spatial transcriptomics of cirrhotic vs. non-cirrhotic HCC tissue
2025	COVID-19 severity prediction using multimodal clinical and imaging data
2023 – 2024	Hypertension-progression risk groups in hyperlipidemia patients (KoGES cohort)
2023	CatchYou: criminal composite sketch generation from witness descriptions
2022	Customs notice analysis and monitoring system for global supply-chain crises

Technical Skills

Programming	Python, R
Frameworks	PyTorch, TensorFlow, Scikit-learn
Tools	Linux, Git, Docker, Nextflow
Research	Machine Learning, Deep Learning, Multimodal Learning
Languages	Korean (Native), English (Fluent)
Certifications	Engineer Information Processing (2023), SQL Developer – SQLD (2023)

Extracurricular Activities

Domestic

Google Developer Student Clubs, Myongji University — *Seoul, S.Korea*

AI Team Member — *Oct. 2022 - Dec. 2023*

- Engaged in machine learning and deep learning study groups
- Collaborated on AI projects and participated in AI competitions

Data Youth Campus, MSIT & KDATA — *Seoul, S.Korea*

Participant (Dongguk University) — *Jul. 2022 - Aug. 2022*

- Completed a hands-on data talent program run by the Ministry of Science and ICT and KDATA, hosted at Dongguk University
- Conducted practical machine learning and data analysis projects

International

Camino de Santiago (French Way) — *Spain*

Pilgrim — *Sep. 2026 - Oct. 2026*

Medical Mission Trip to the Philippines (GMMA) — *Manila, Philippines*

Volunteer — *Jan. 2024*

- Participated in medical volunteer activities with international healthcare professionals

International Work Camp in Iceland — *Reykjavik, Iceland*

Volunteer — *Dec. 2019 - Jan. 2020*

- Participated in environmental campaigns and intercultural exchange with international volunteers

Mission Trip to Vietnam — *Hanoi, Vietnam*

Volunteer — *Aug. 2018*

- Participated in intercultural exchange and community volunteer activities